

Team

2: Main + Company

Team 2 — Main + Company

You build **main.c** (program entry), **company.h**, and **company.c** (Company panel). You wire the whole app together and give dealers/retailers a front door via the main menu.

Your deliverable: main.c starts the app; company.c lets Company view dealers, products, and warehouse stock.

Team	Files	Role
Team 1	data.c, data.h, display.c, display.h	Database + UI (YOU depend on this)
Team 2 (YOU)	main.c, company.c, company.h	Start program + Company panel
Team 3	dealer.c, dealer.h, retailer.c, retailer.h	Dealer + Retailer + orders

Three-team split

1. What You Are Building

1.1 main.c

- Calls **init_data()** from Team 1.
- Shows main menu in a **do-while** loop.
- Routes choice 1 → `company_menu()`, 2 → `dealer_menu()`, 3 → `retailer_menu()`.

1.2 company.c

- Password login (`verify_password` from Team 1).
- View-only menus: list dealers, list products, view company stock.
- No ordering — Company only monitors.

2. What You Need From Teammates

From Team 1 (required first)

File / symbol	Why
data.h + data.c	init_data(), product_total, dealer arrays
display.h + display.c	print_menu_*, print_error, verify_password
init_data()	Call at start of main()

Team 1 dependencies

From Team 3 (for full main.c)

You need **dealer_menu()** and **retailer_menu()** declared in dealer.h and retailer.h. Until Team 3 finishes, use **stubs**:

c

```
/* dealer.h stub while Team 3 works */
void dealer_menu();

/* temporary dealer.c stub */
void dealer_menu() {
    printf("Dealer panel coming soon (Team 3).\n");
}
```

3. Build Order

1. Get Team 1's four files (data + display).
2. Create **company.h** with void company_menu();
3. Build **company.c** — list functions then company_menu().
4. Build **main.c** with company only first.
5. Add dealer/retailer includes when Team 3 delivers headers.
6. Full integration compile with all 6 .c files.

4. Step-by-Step: company.h

C

```
void company_menu();
```

5. Step-by-Step: company.c

Step 5.1 — Includes

C

```
#include <stdio.h>
#include "company.h"
#include "data.h"
#include "display.h"
```

Step 5.2 — list_dealers()

C

```
void list_dealers() {
    int i;
    print_subtitle("All Dealers");
    print_table_dealer_header();
    for (i = 0; i < dealer_total; i++) {
        print_table_dealer_row(dealer_id[i], dealer_name[i]);
    }
    printf("+-----+-----+\n");
}
```

Step 5.3 — list_products() and show_company_stock()

Both loop over product_total and print product_id, product_name, company_stock. list_products uses subtitle "All Products"; show_company_stock uses "Company Warehouse Stock".

C

```
void list_products() {
    int i;
    print_subtitle("All Products");
    print_table_product_header();
    for (i = 0; i < product_total; i++) {
        print_table_product_row(product_id[i], product_name[i], company_stock[i]);
    }
    print_table_footer();
}
```

Step 5.4 — company_menu()

C

```
void company_menu() {
    int choice;
    print_subtitle("Company Login");
    if (!verify_password()) return;
    print_success("Login successful.");

    do {
        print_menu_header("COMPANY PANEL");
        print_menu_item(1, "View All Dealers");
        print_menu_item(2, "View All Products");
        print_menu_item(3, "View Company Stock");
        print_menu_item(0, "Back to Main Menu");
        printf("\nEnter your choice: ");
        scanf("%d", &choice);

        if (choice == 1) list_dealers();
        else if (choice == 2) list_products();
        else if (choice == 3) show_company_stock();
        else if (choice != 0) print_error("Invalid choice.");
    } while (choice != 0);
}
```


6. Step-by-Step: main.c

Step 6.1 — Includes

C

```
#include <stdio.h>
#include "data.h"
#include "display.h"
#include "company.h"
#include "dealer.h"
#include "retailer.h"
```

Step 6.2 — main() body

C

```
int main() {
    int choice;
    init_data();

    do {
        print_menu_header("DISTRIBUTION MANAGEMENT SYSTEM");
        print_menu_item(1, "Company Login");
        print_menu_item(2, "Dealer Login");
        print_menu_item(3, "Retailer Login");
        print_menu_item(0, "Exit");
        printf("\nEnter your choice: ");
        scanf("%d", &choice);

        if (choice == 1) company_menu();
        else if (choice == 2) dealer_menu();
        else if (choice == 3) retailer_menu();
        else if (choice != 0) print_error("Invalid choice.");
    } while (choice != 0);

    printf("\nThank you for using DMS. Goodbye!\n\n");
    return 0;
}
```

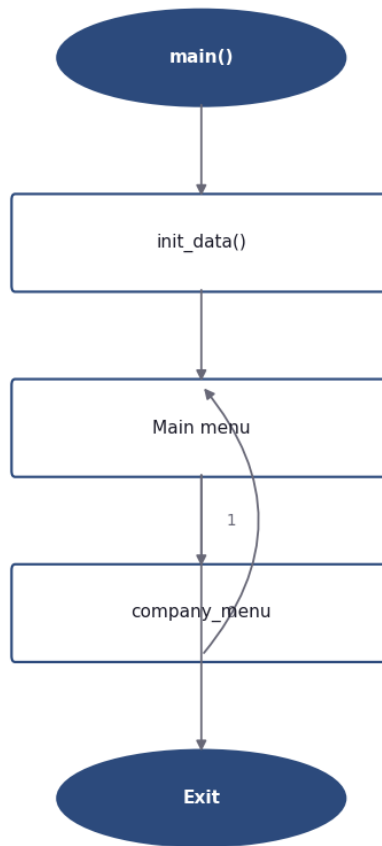


Figure 1: Your main.c flow (Company path)

7. Test Your Part

Phase A — Company only (before Team 3)

BASH

```
gcc main.c data.c display.c company.c dealer_stub.c retailer_stub.c -o dms
./dms
```

Use stub files with empty dealer_menu() and retailer_menu(). Test: main → 1 → password 1122 → options 1,2,3 → 0.

Phase B — Full app

BASH

```
gcc main.c data.c display.c company.c dealer.c retailer.c -o dms
./dms
```

Test	Steps	Expected
Company dealers	1 → 1122 → 1	Dhaka + Chittagong table
Company products	1 → 1122 → 2	Rice, Oil, Sugar
Company stock	1 → 1122 → 3	Same with warehouse label
Exit	0	Goodbye message

Company panel tests

8. Integration With Team 3

When Team 3 adds real dealer.c and retailer.c, remove stubs. Your main.c should not change — only swap object files at link time.

After a dealer order, Company option 3 should show lower company_stock — proves Team 1's shared arrays work.

Team 2 checklist

- main.c calls init_data() before any menu.
- company_menu() uses verify_password and do-while.
- All three company views use for loops over totals from data.h.
- Full gcc command links all 6 implementation .c files.
- Commit: main.c, company.c, company.h.